

Power Carriers

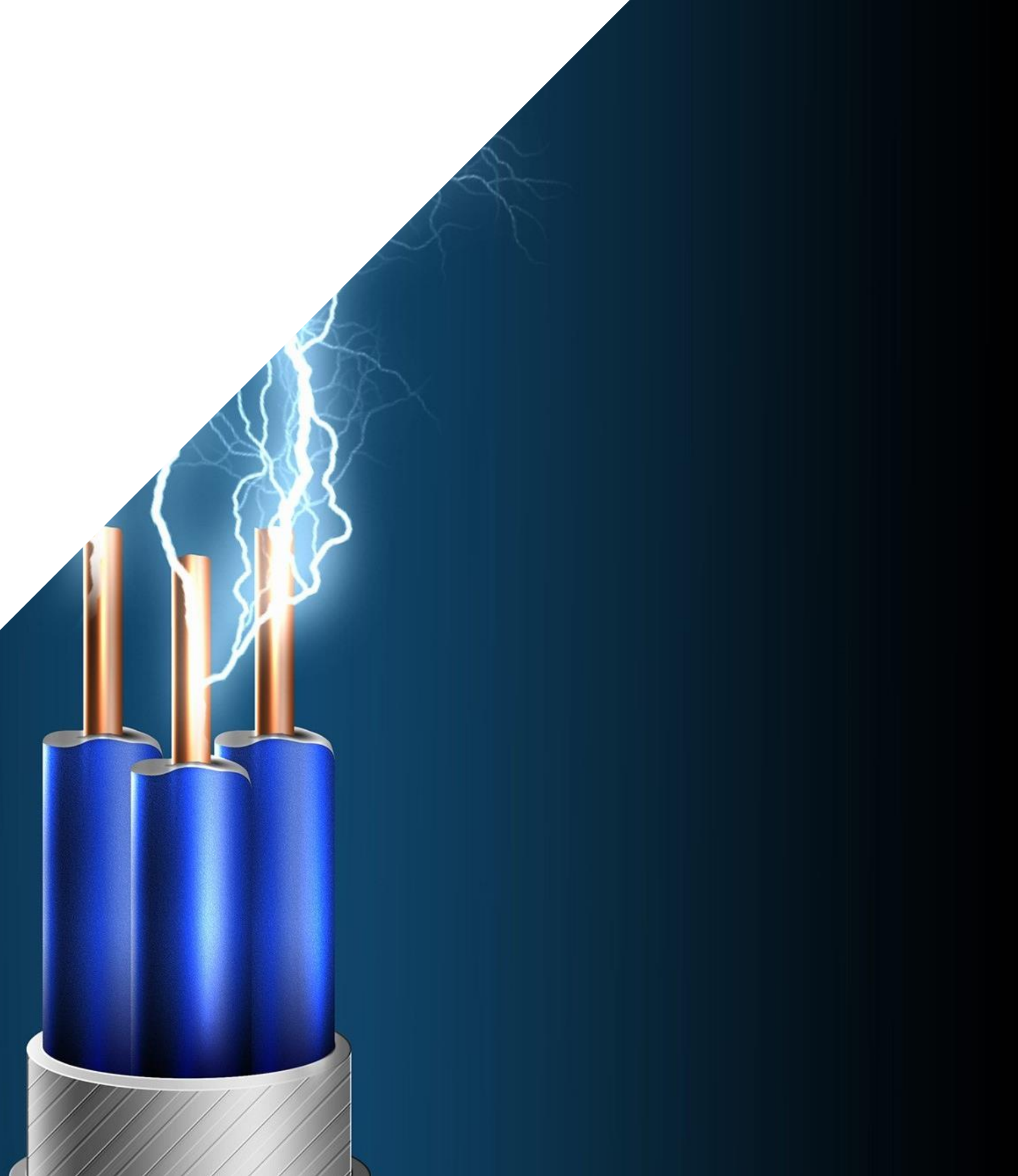
Energy Transmission
2026-03-30

Felix Feustel

Contents

- ◆ Overhead Transmission Lines
- ◆ Underground Cables
- ◆ Exercise:
 - ◆ Charging current of underground cable

Introduction



Example: Power carriers in Dutch power system

Voltage range [kV]	Aboveground [km]	Underground [km]
220 / 380	2719	16
50 / 110 / 150	5580	3614
3–25	—	105599
0.4	174	150623

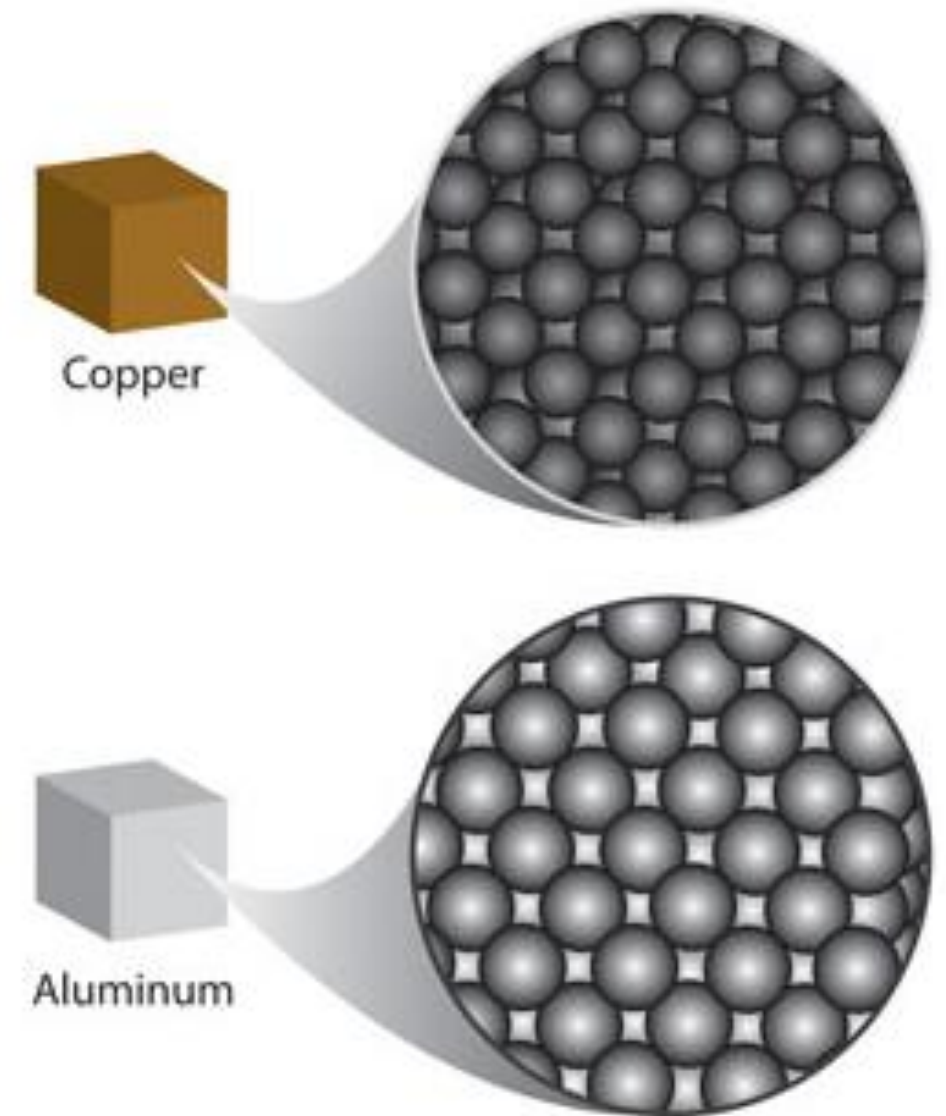
- ♦ High-voltage level: Overground preferred
- ♦ Low-voltage level: Underground preferred

Advantages and disadvantages

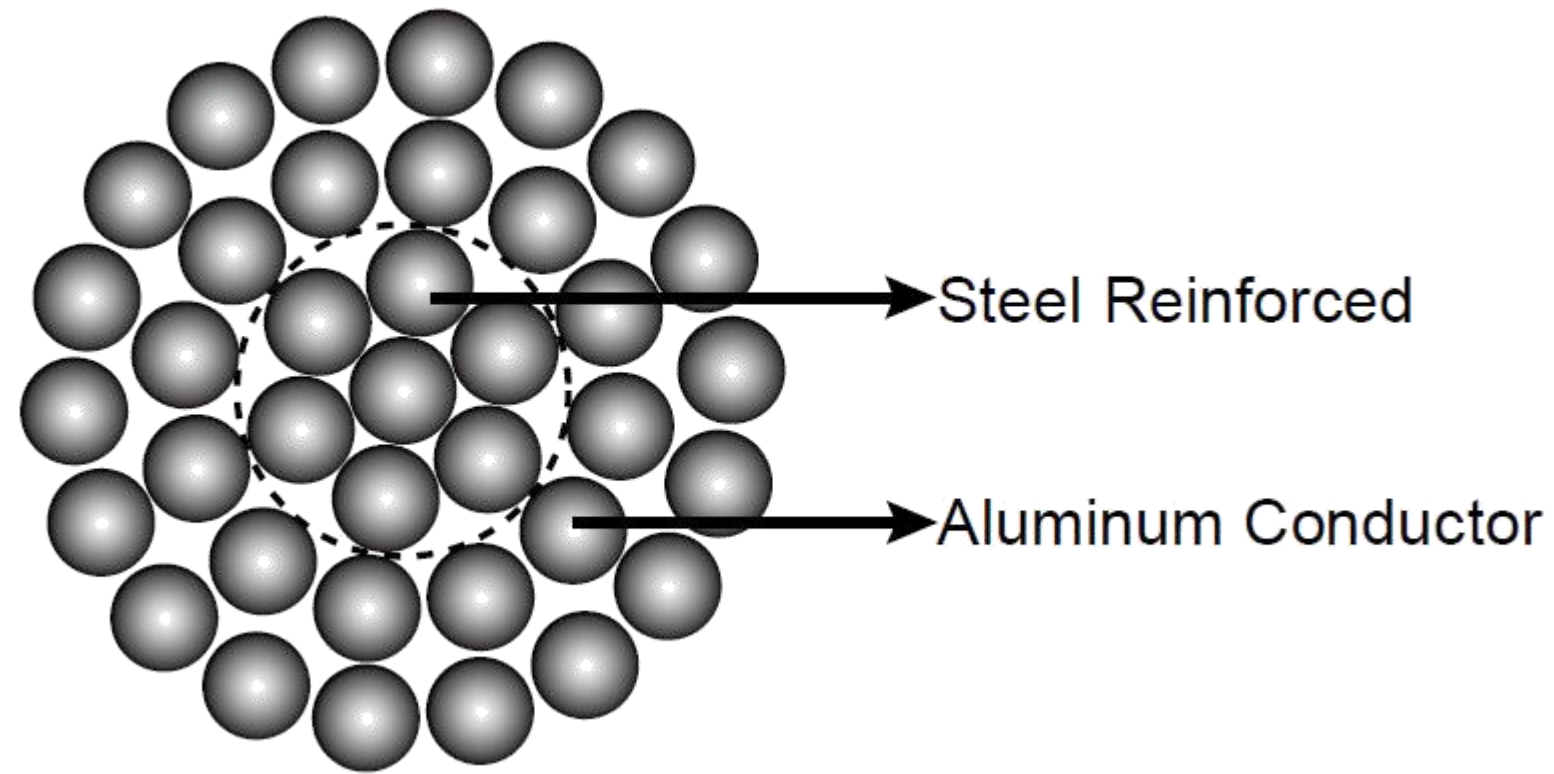
- ♦ Economical and environmental reasons to choose one over the other
- ♦ Cables:
 - Advantages: Reliability, fit for urban areas, less maintenance
 - Disadvantage: Higher costs (8.2 more expensive)
 - Durability: 30-40 years
- ♦ Overhead lines:
 - (Arguments apply in reverse)
 - Durability: 100-120 years

Copper and Aluminium

- ◆ Density
 - Copper: 8900 kg/m^3
 - Aluminum: 2700 kg/m^3
 - ◆ Resistivity
 - Copper: $18 \times 10^{-9} \Omega\text{m}$
 - Aluminium: $29 \times 10^{-9} \Omega\text{m}$
- Aluminium conductor has half the weight of a copper conductor with the same resistance
- Preferred conductor material for overhead lines – but, on its own, it has low tensile strength



Aluminum conductor with steel core

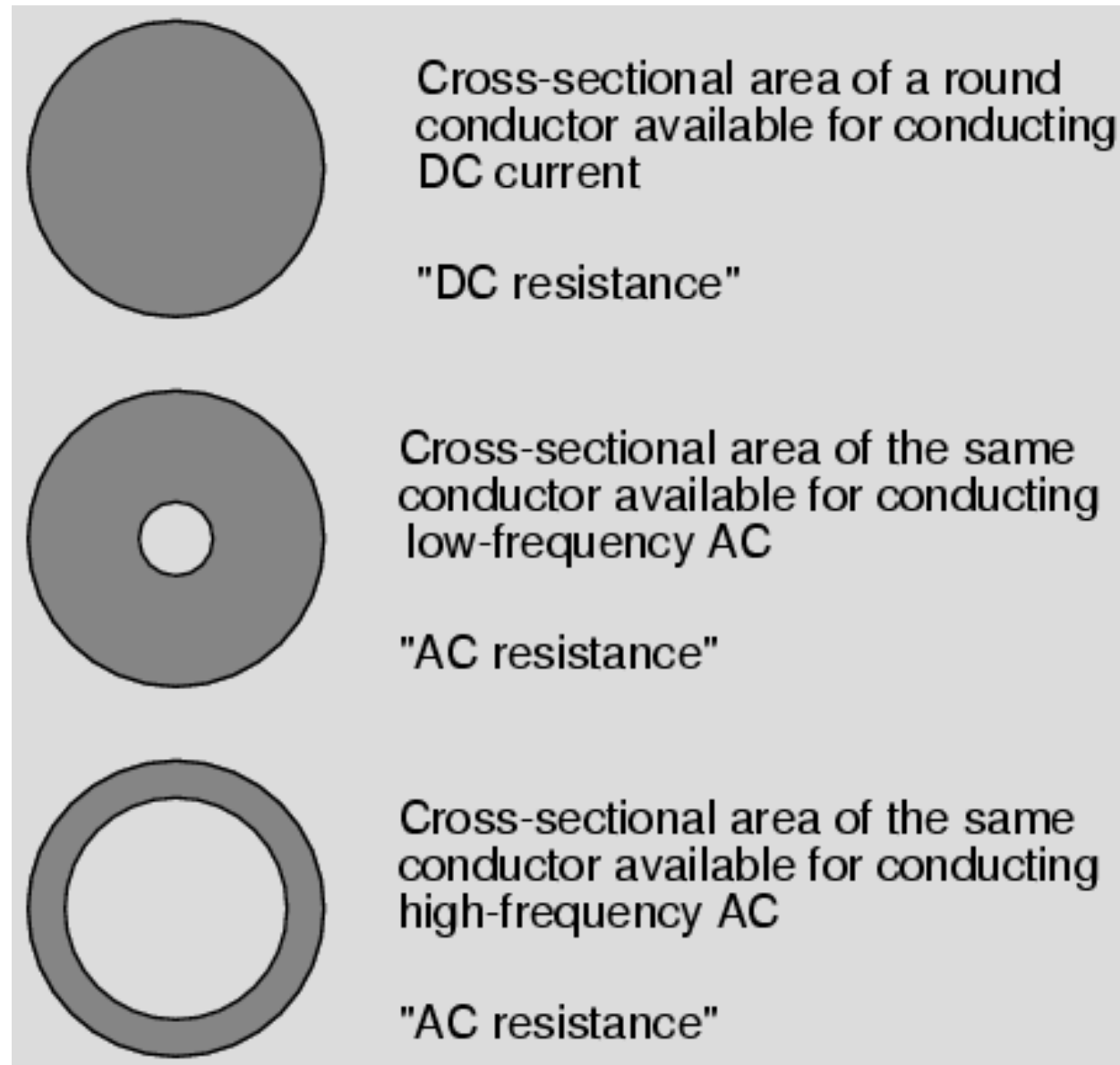


- ◆ Example: 27 Aluminum strands, 7 steel strands
- ◆ Question: Does steel add to the resistance of the wire?



Skin effect

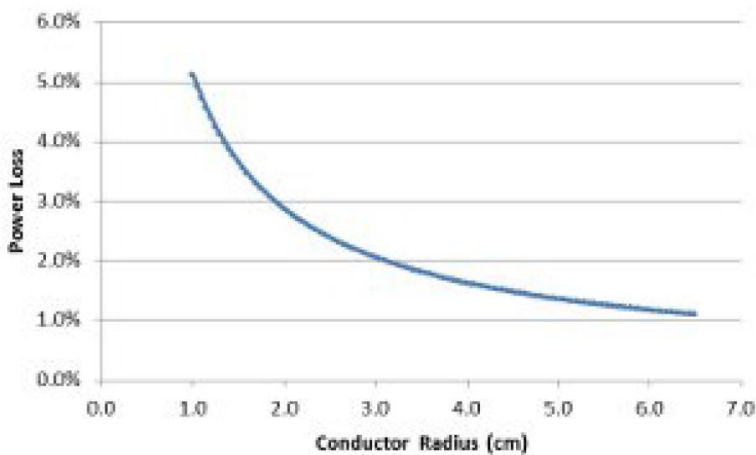
Tendency of AC current to become distributed within a conductor such that the current density is greater at the surface of the conductor than at its core



- ♦ Steel core (centered in the middle) has neglectible impact on resistance
- ♦ But: AC resistance higher than DC resistance due to Skin effect (typically 1...6 % at 50 Hz)
→ Skin losses

Example: Resistive losses

Parameter		Case 1	Case 2	Case 3
d	Line Separation	10 m		
a	Conductor Radius	0.015m		
L	Inductance Per Meter	2.6μH/m		
f	Frequency	60 Hz	50 Hz	60 Hz
σ	Conductivity of the Metal	3.82×10^7 S/m (Al)	3.82×10^7 S/m	6.17×10^7 S/m (Ag)
I _B	Bessel Correction Factor	1.1	1.1	1.1
δ	Skin Depth	10.5 mm	11.5 mm	8.3 mm
R _l	Resistance per Meter	29.1 μΩ/m	26.5 μΩ/m	22.9 μΩ/m
α	attenuation factor	18.6×10^{-9} /m	17.0×10^{-9} /m	14.7×10^{-9} /m
μ ₀	Permeability of Free Space	$4\pi \times 10^{-7}$ H/m		
c	Speed of Light	3×10^8 m/sec		
%P _{Rloss} (1km)		37.2 ppm	34.0 ppm	29.3ppm
%P _{Rloss} (1000km)		3.66%	3.34%	2.89%



Skin depth (90% of power is carried above this distance from surface):

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}}$$

Resistance per meter:

$$R_l = \frac{I_B}{2\pi a \sigma \delta}$$

Inductance per meter:

$$L = \frac{\mu}{\pi} \ln\left(\frac{d}{a}\right)$$

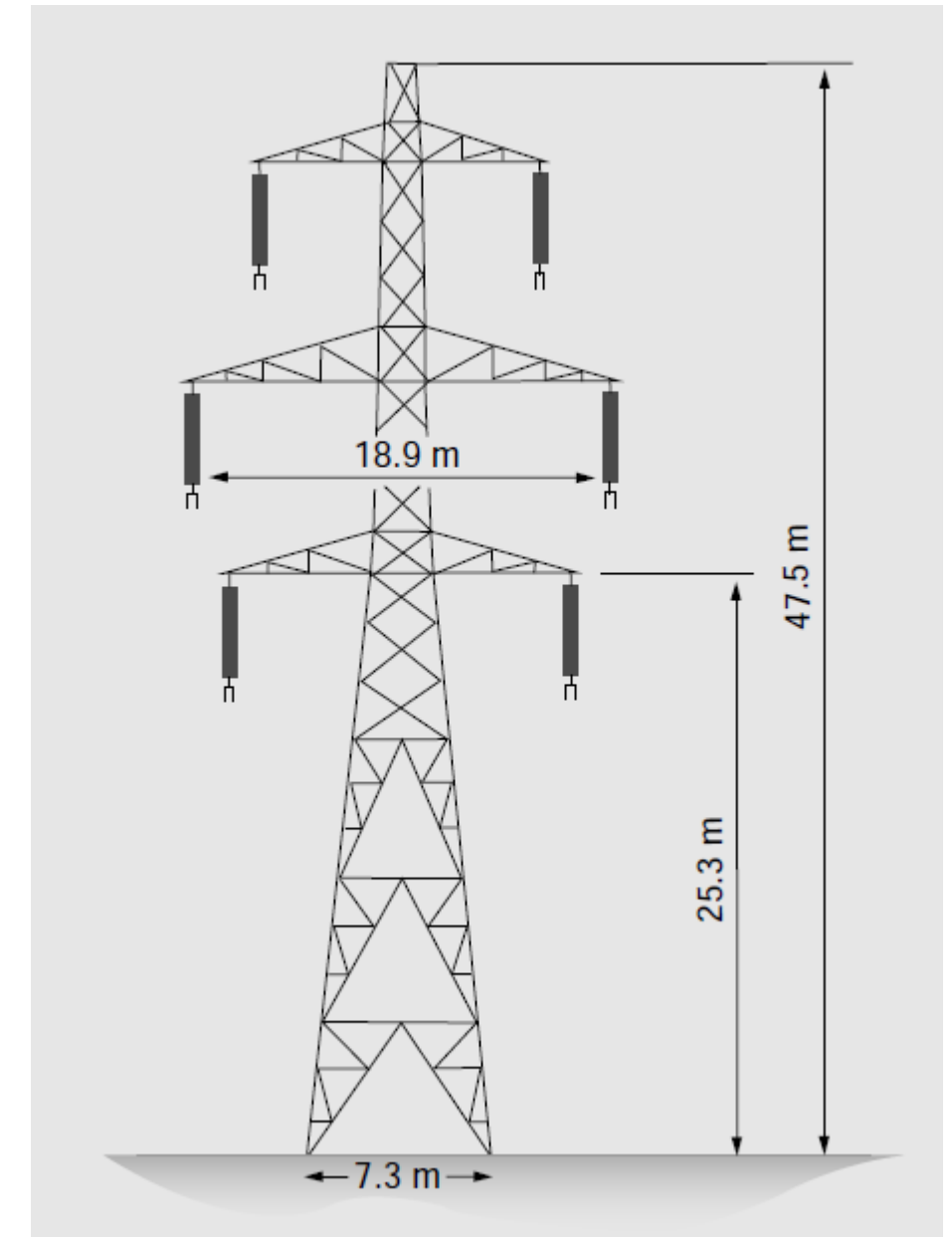
(Harting 2010)

Overhead lines

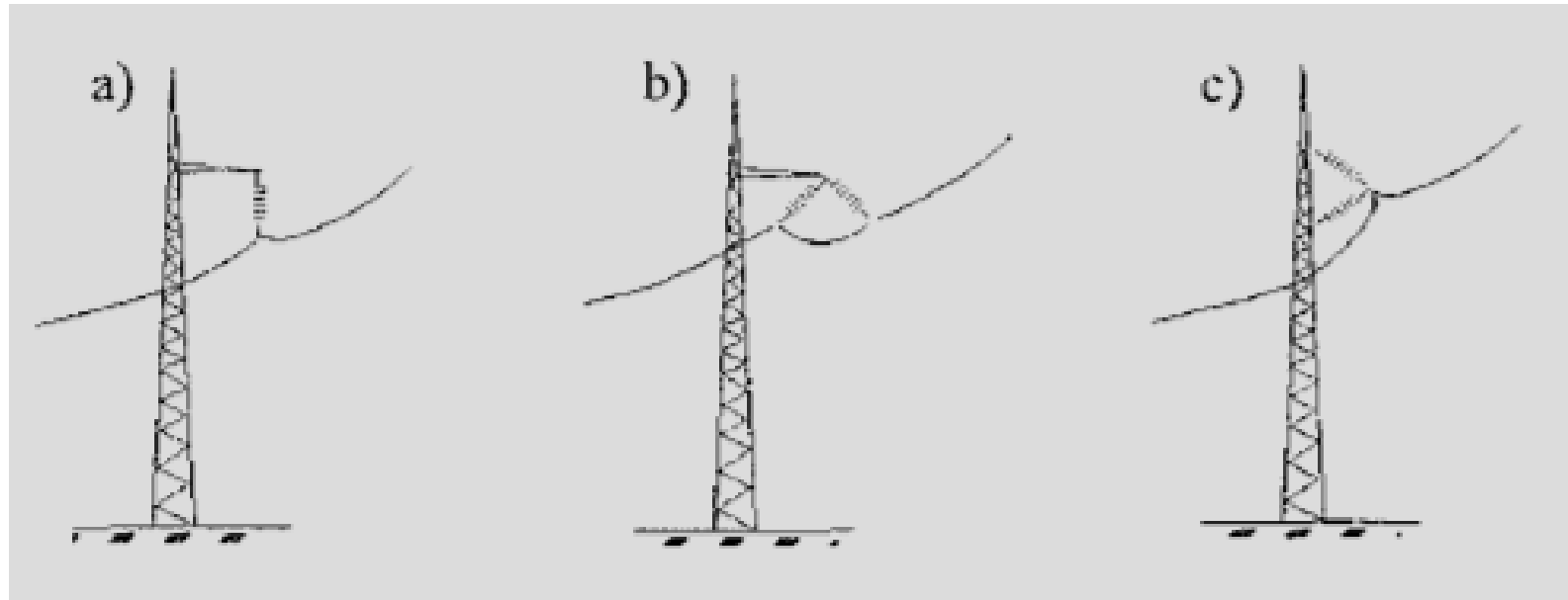


Characteristics

- ◆ Supported by transmission line poles or towers
- ◆ Surrounding air serves as insulation medium
- ◆ Question: What determines tower height?
 - Minimum clearance from ground (depends on voltage level)
 - Sag of conductors
 - Span (distance) between towers



Pole types



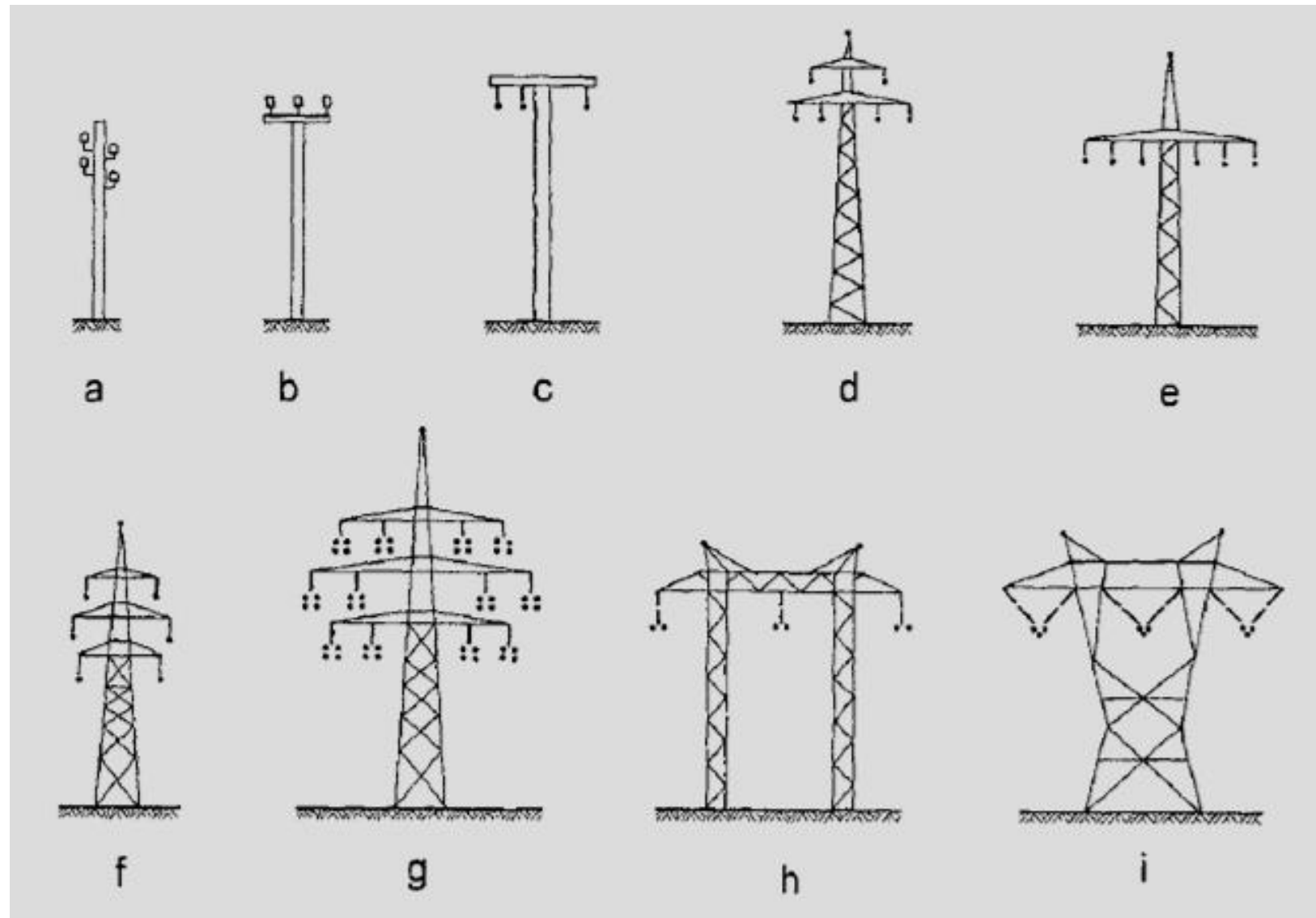
Legend:

a) supporting post b) anchor pylon c) deviation tower

right) termination pylon



Mast construction types

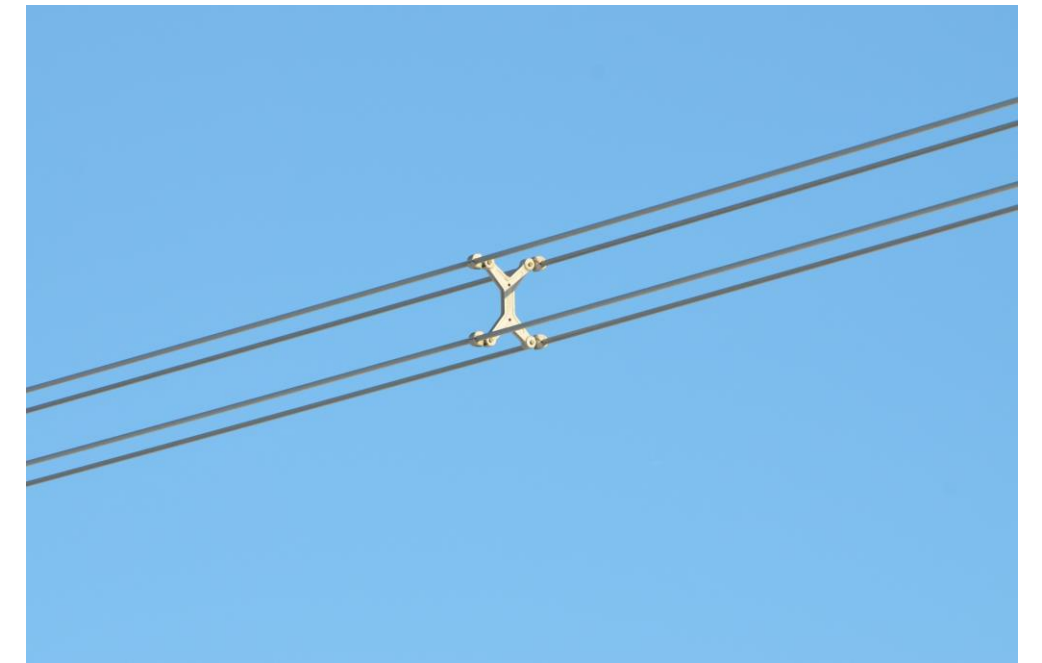


Legend:

a) b) wooden (LV), c) concrete (MV), d) Donau type, e) single-level type, f) British type, g) bundle conductor, h) portal type i) Y type (all HV)

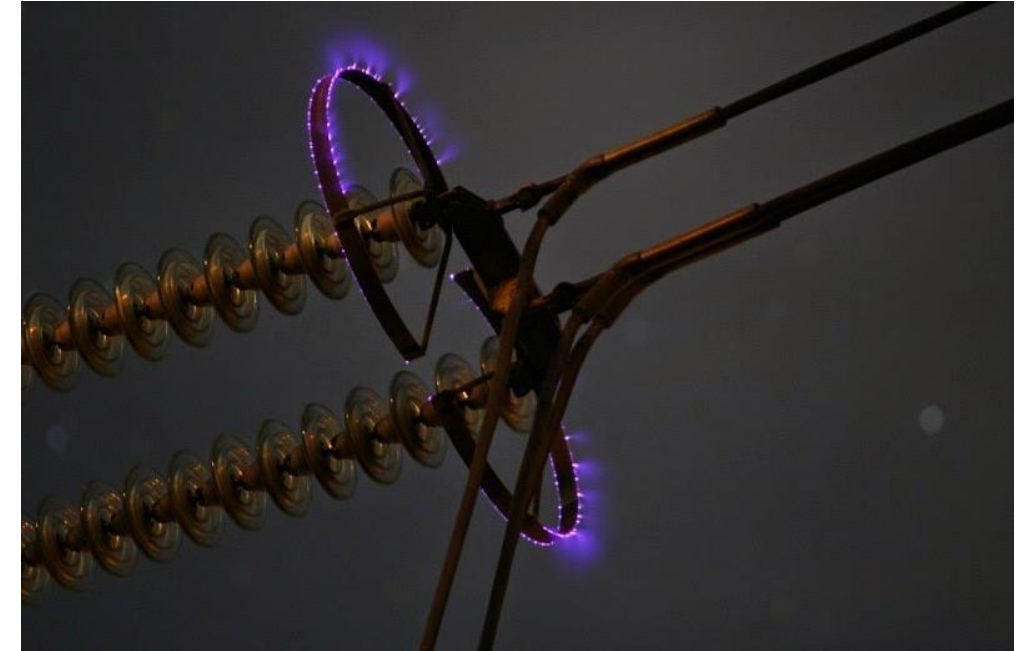
Bundle conductors

- ◆ Used above 230 kV
- ◆ Separated by means of spacers at regular intervals
- ◆ Reduce corona losses (they are geometry-dependent)



Corona losses

- ♦ Electric field near a conductor can be sufficiently concentrated to ionize air
- ♦ Can result in a partial discharge of electrical energy
- ♦ Utility companies try to reduce the amount of corona
 - Power losses
 - Can damage system components over time
 - Radio noise



Factors influencing corona losses

- ◆ During wet and humid conditions, water drops collect on the conductors and increase corona activity (crackling sound)
- ◆ Contributing factors:
 - voltage of line
 - diameter of conductors
 - location of conductors in relation to each other
 - elevation above sea level
 - condition of the conductors
 - local weather conditions
- ◆ Not dependent on load flow

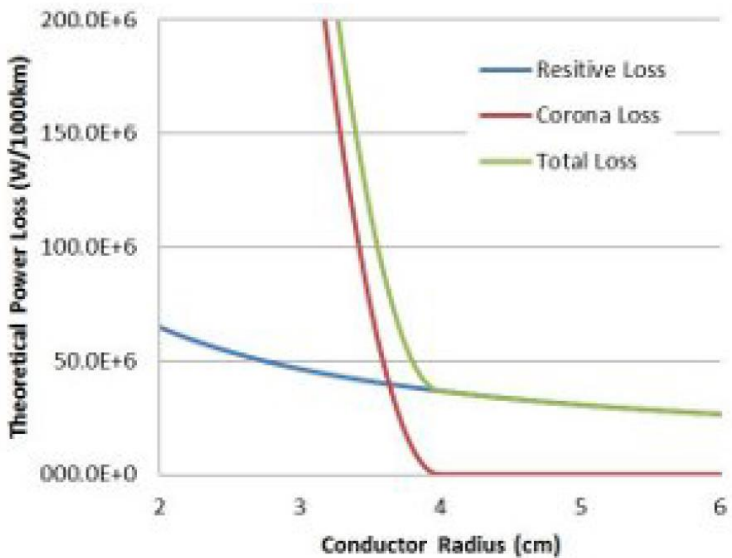
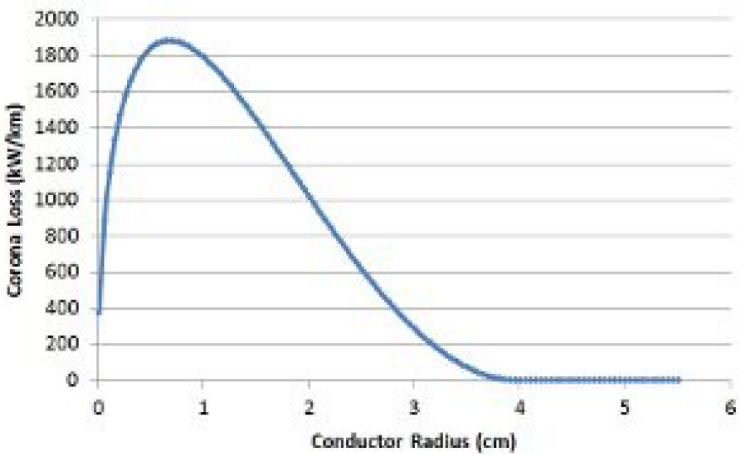


Example: Corona losses

Parameter		Sample Value
k_0	Fixed constant	241
g_0	Disruptive Gradient in Air	21.1 kv/cm
k_d	Normalized Air Density Factor 1 (25 °C, 76 cm pressure)	1
a	Radius of Conductor	3.5 cm (see Fig. 2)
d	Conductor Spacing	1000 cm
f	Frequency	60 Hz
k_i	Wire Irregularity Factor	0.95 (weathered wires)
V_0	Line Voltage to Neutral (1/1.73 x Voltage Between Conductors)	442 kV (765 kV/1.73)
	Disruptive Critical Voltage ($g_0 k_i a k_d \ln(d/a)$)	397 kv
	Corona Loss kW/km/line	25kW/km
	Corona Loss % (1000km line at 2.25 GW)	3.3%

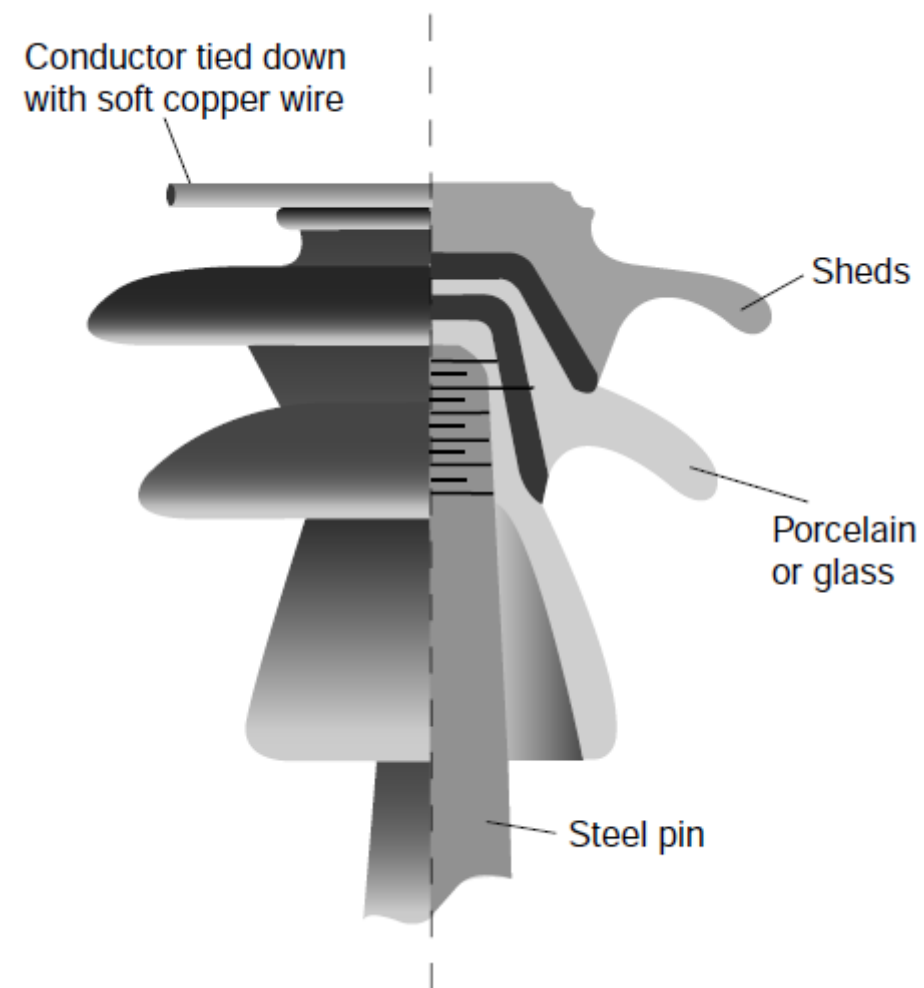
Total amount of power loss in a wire due to corona effect [Peek 1911]:

$$P = \frac{k_0}{k_d} (f + 25) \sqrt{\frac{a}{d}} [V_0 - g_0 k_i a k_d \ln(\frac{d}{a})]^2 \times 10^{-5} \text{ kW/km}$$

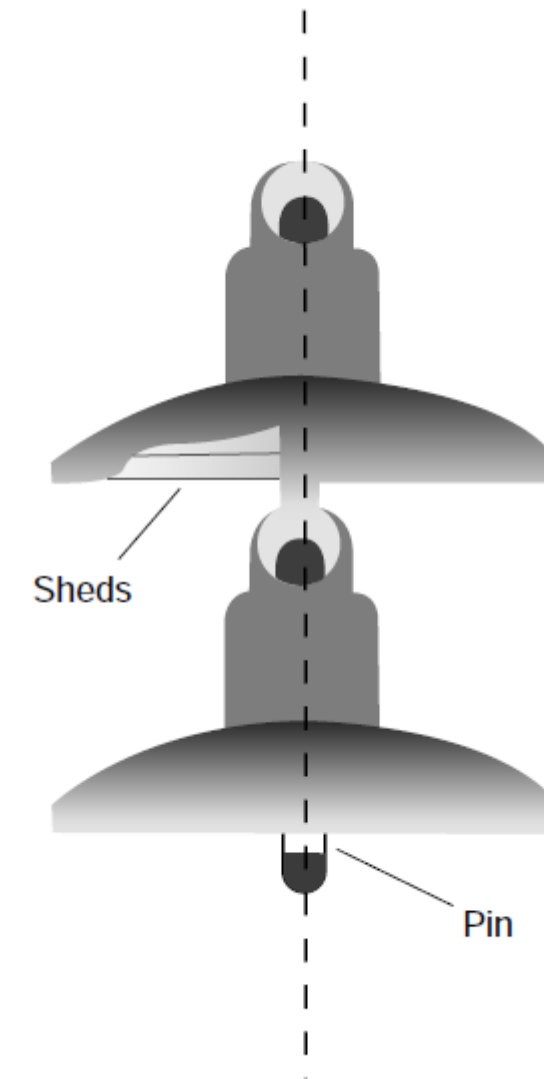


(Harting 2010)

Insulators

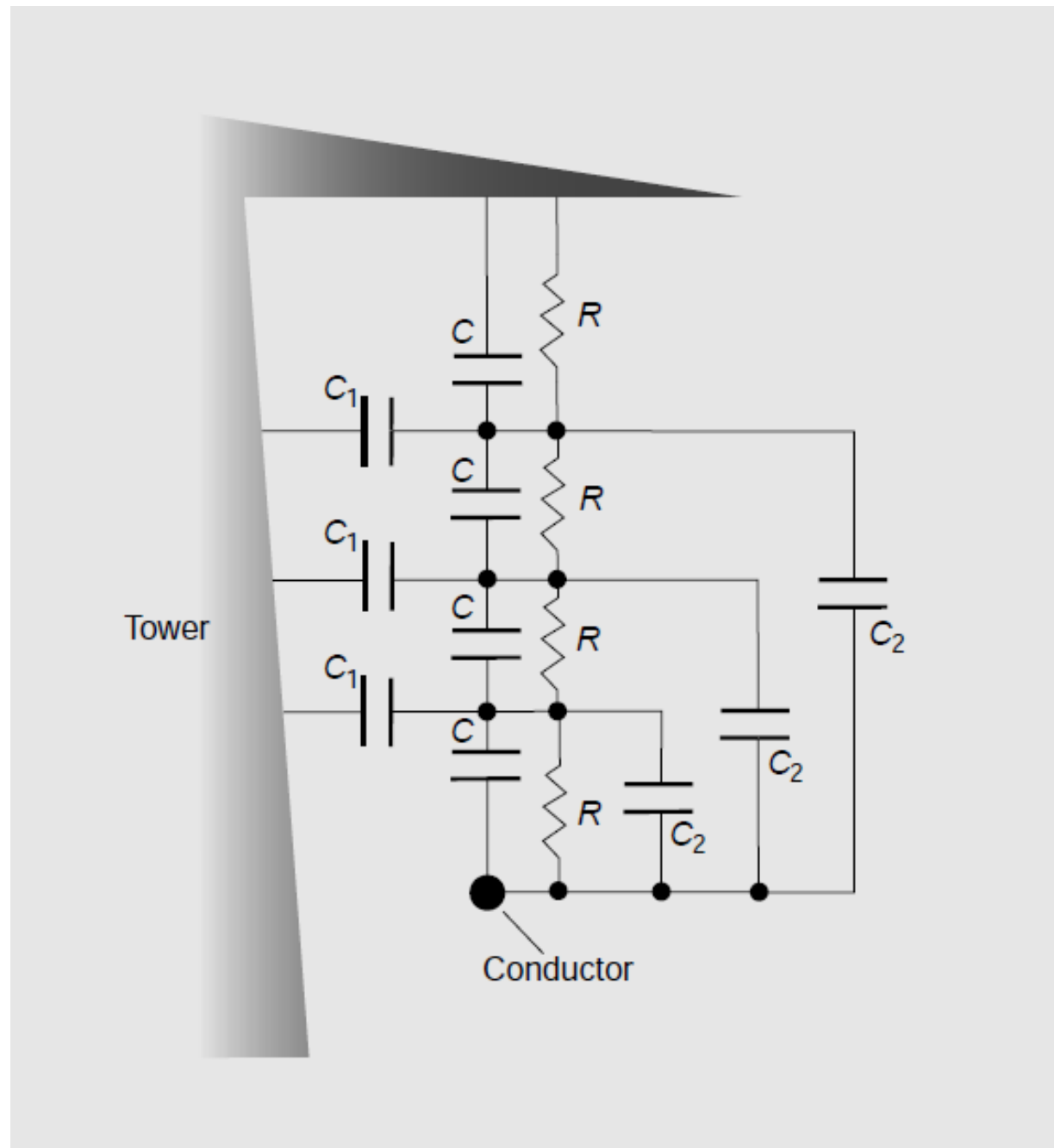


Suspension type



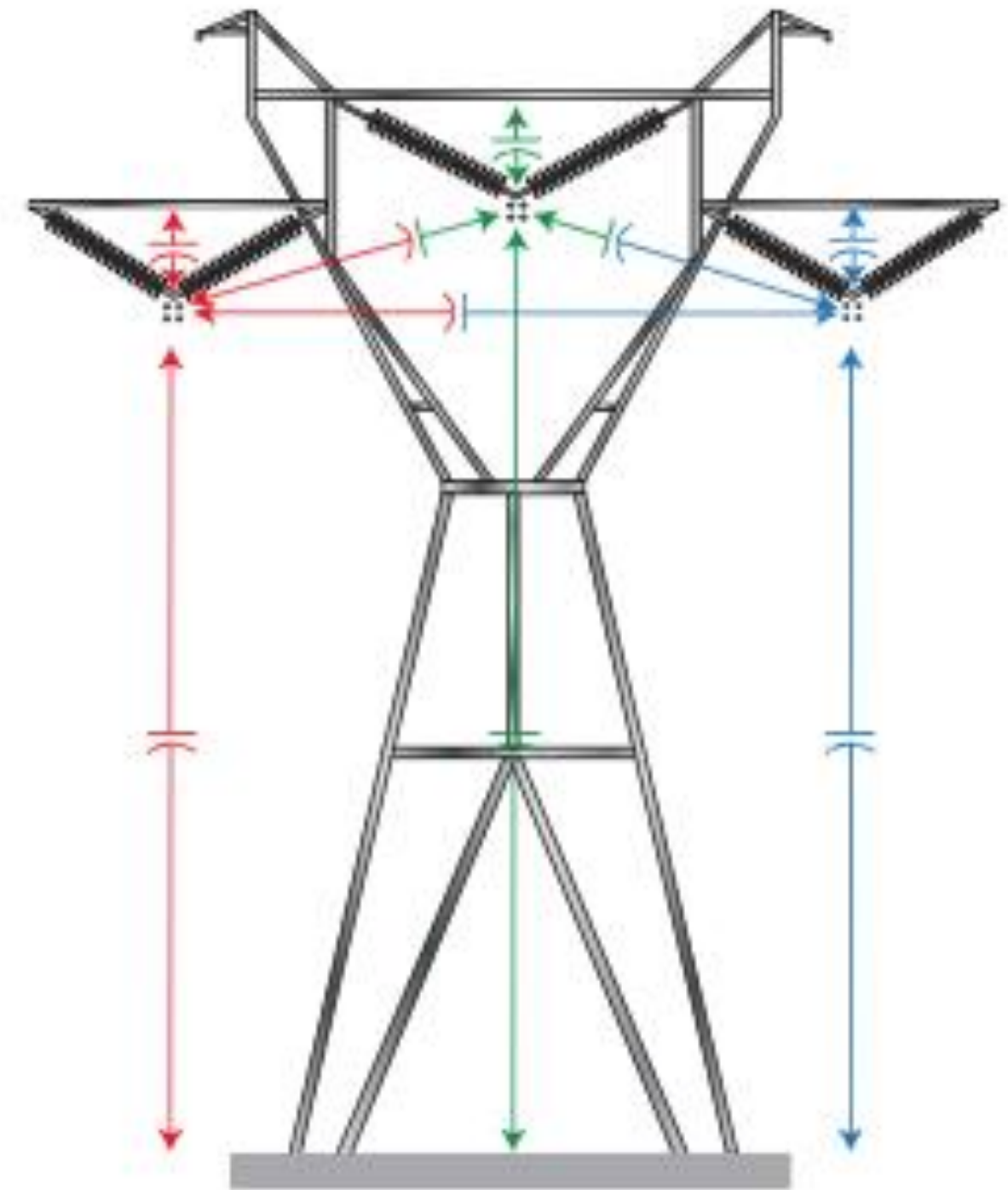
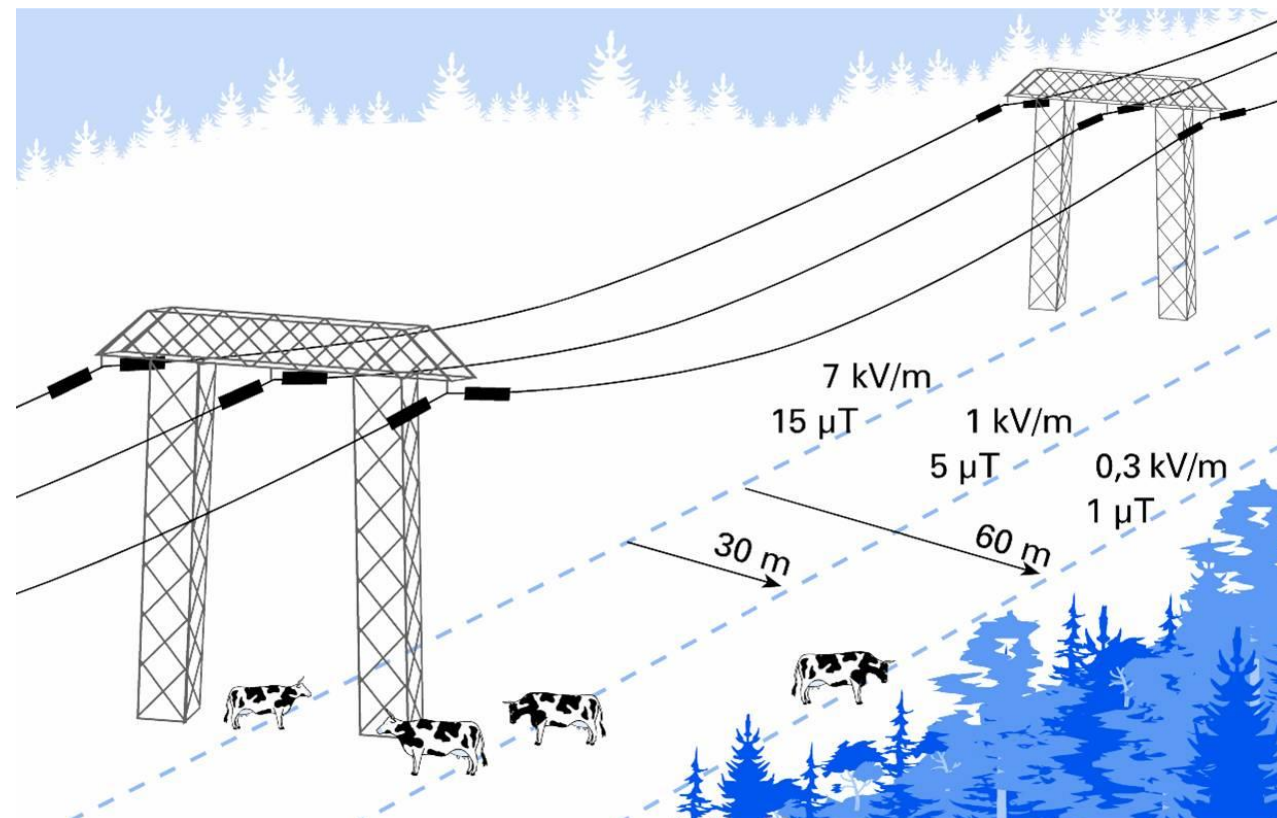
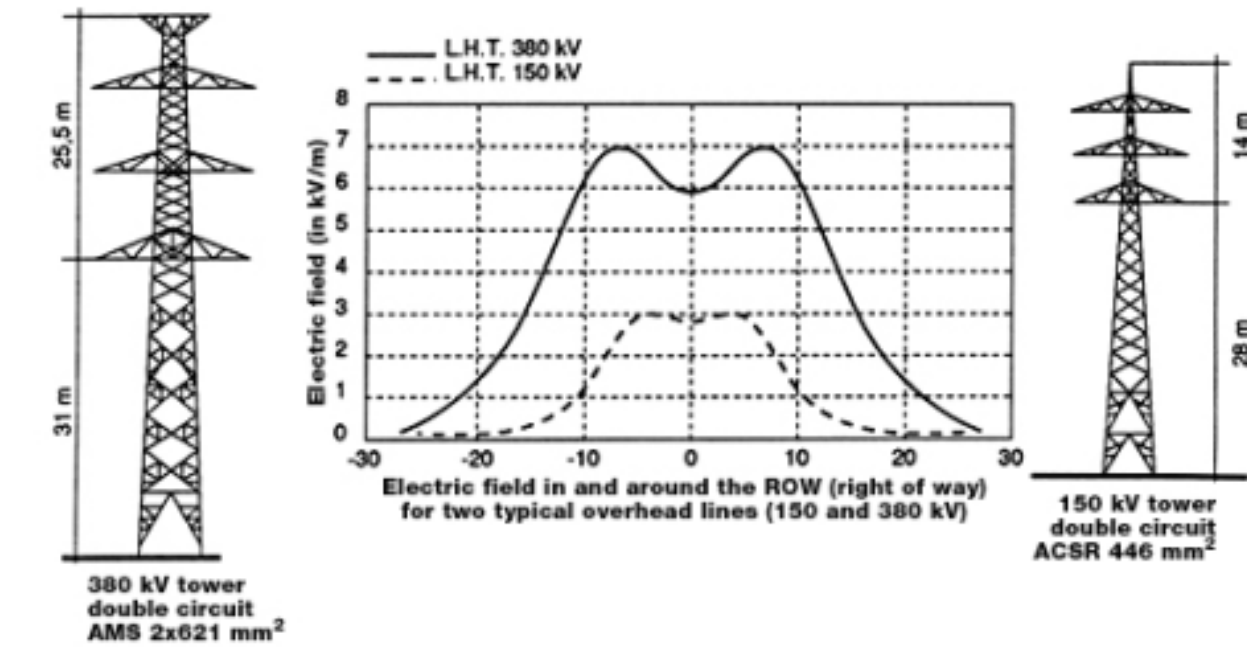
Pin type

Equivalent circuit of four suspension insulators



- ♦ C = self capacitance of discs
- ♦ C_1 = capacitance disc to earth
- ♦ C_2 = capacitance disc to line
- ♦ R = leakage resistance

Electric field and capacitance model



Earth wires

- ◆ Protect overhead lines against lightning strikes

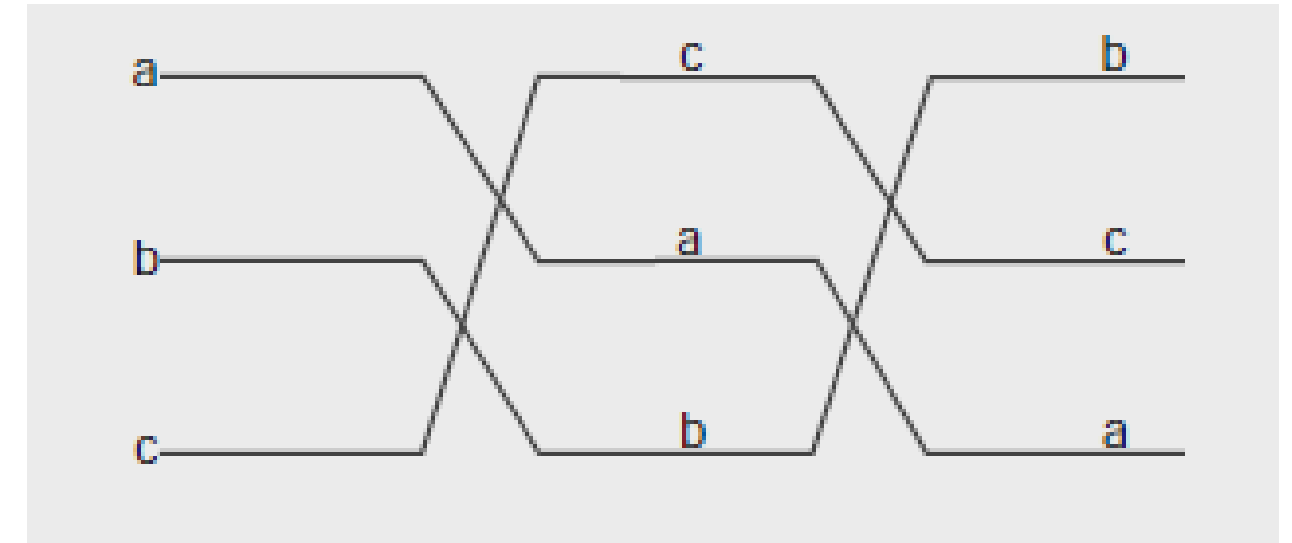


- ◆ May include optical fibers for internal and third-party telecommunication purposes

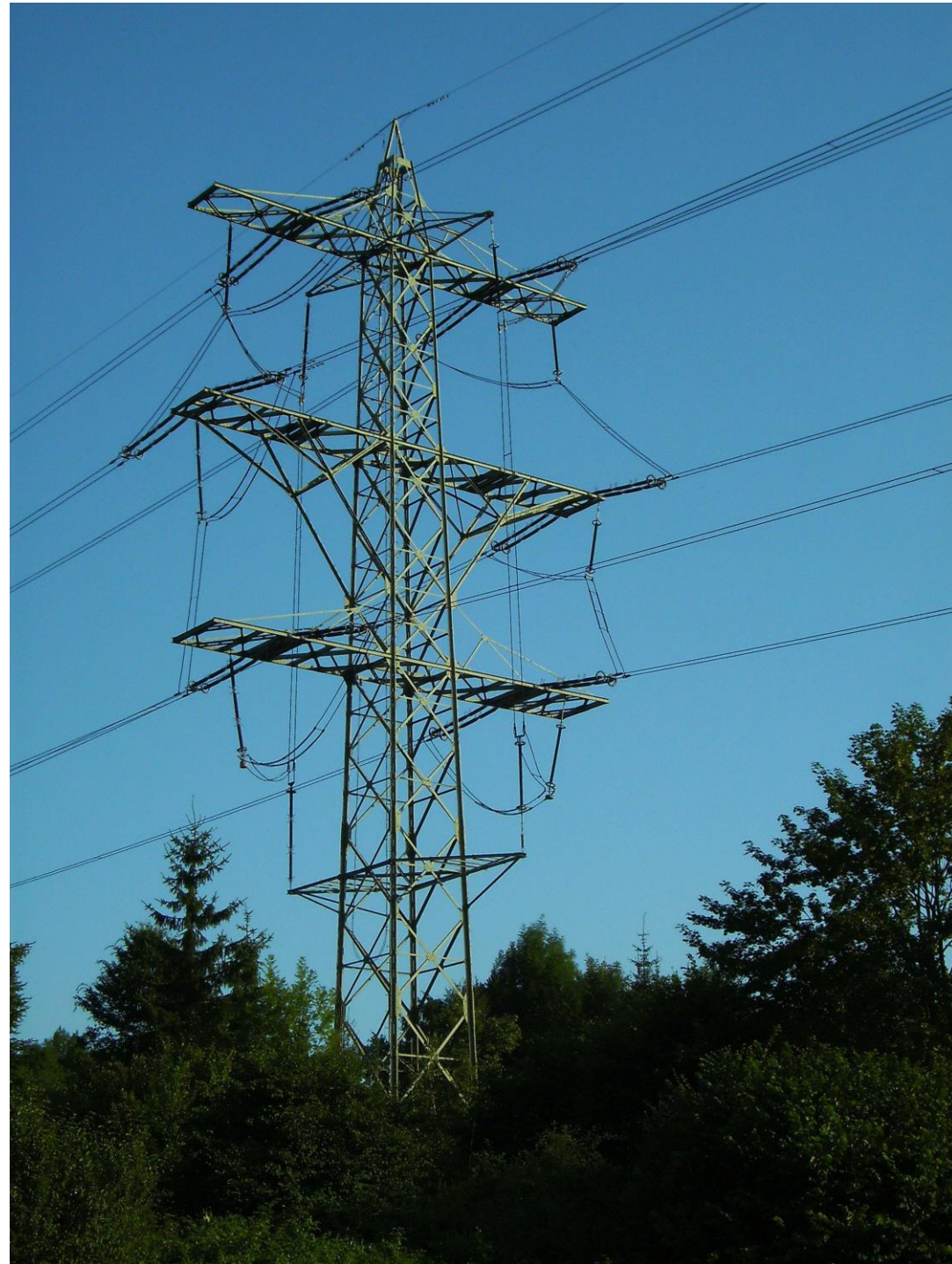


Transposition

- ◆ Slightly different electrical field for each position (due to e.g. line length, distance to ground / earth wire)
- ◆ Rotation of conductors to occupy the next physical position in a regular sequence
- ◆ Transposition is achieved using twisting pylons

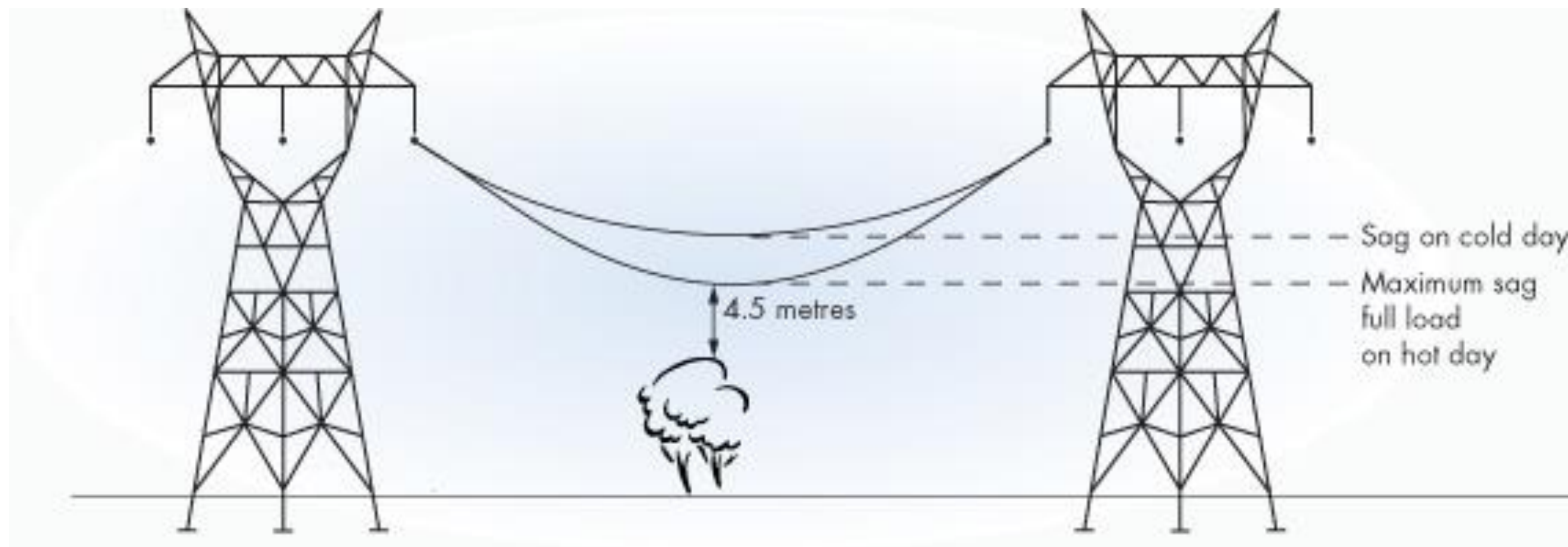


Transposition tower / twisting pylon



Power line sag

- ◆ Overhead conductors elongate with time, temperature, and tension, thereby changing their original positions after installation.
- ◆ Vegetation management necessary
- ◆ Load flow may be optimized considering outside temperature



Galloping lines

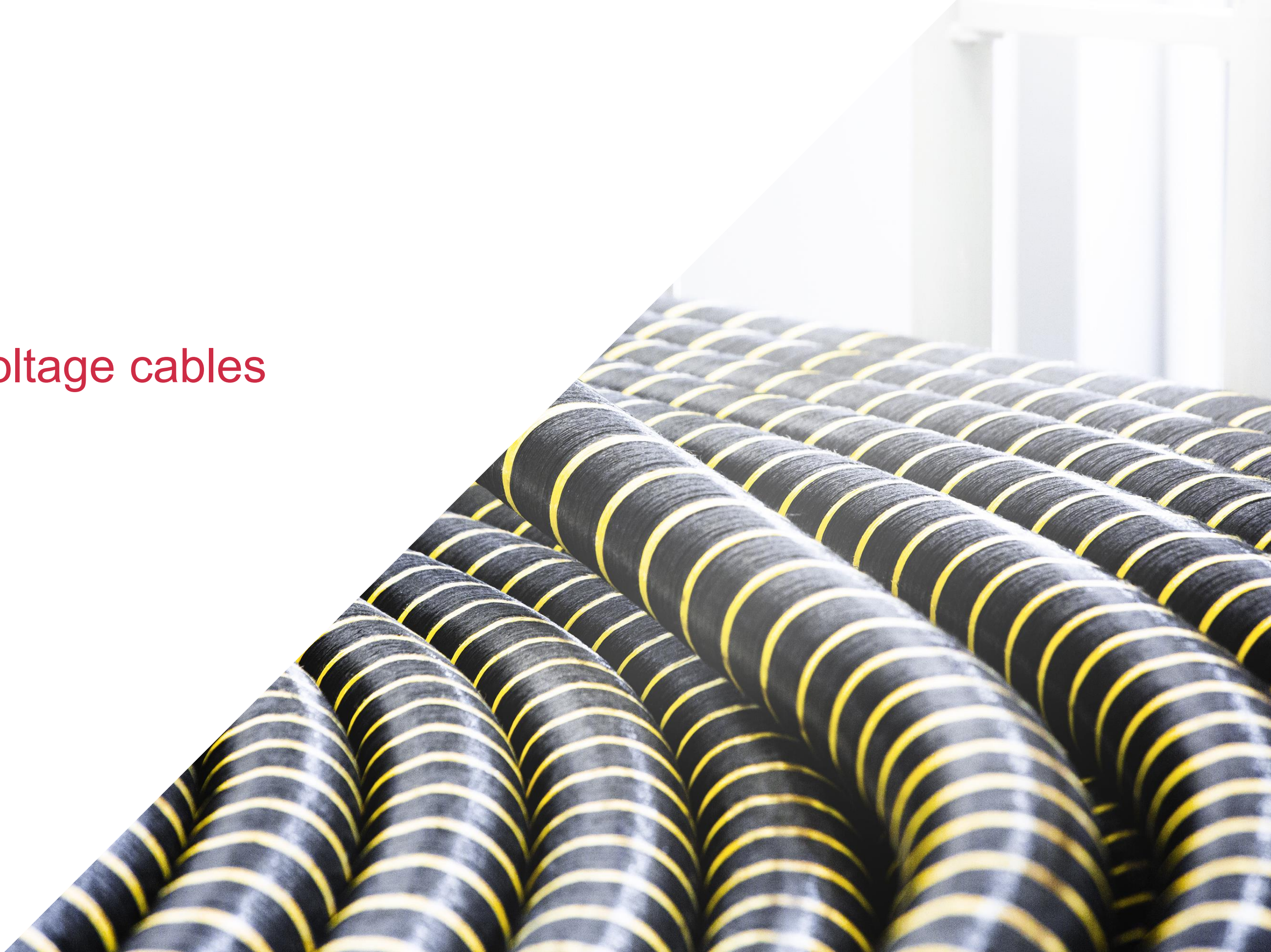
- ♦ High-amplitude, low-frequency oscillation of overhead power lines due to wind
- ♦ Typical frequency: 1 Hz
- ♦ Amplitude typically may exceed one meter (may both occur in horizontal or vertical plane)
- ♦ Danger of flashover
- ♦ Stress on insulators and pylons



Dampers



High-voltage cables



Schematic of 150 kV cable

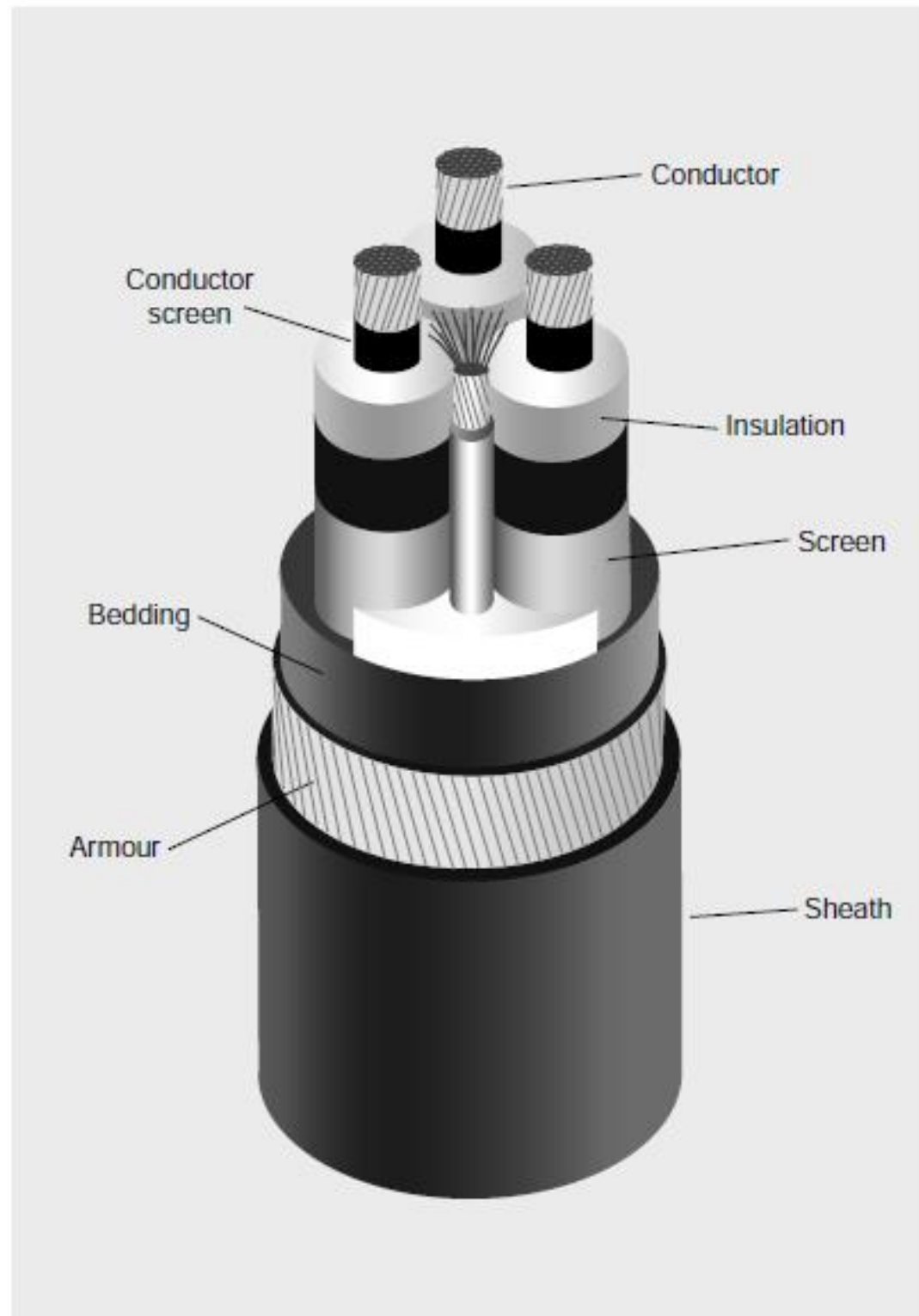


Figure 3.30 150kV XLPE insulated 3 core cable with fibre optic communications

10 kV and 380 kV cable



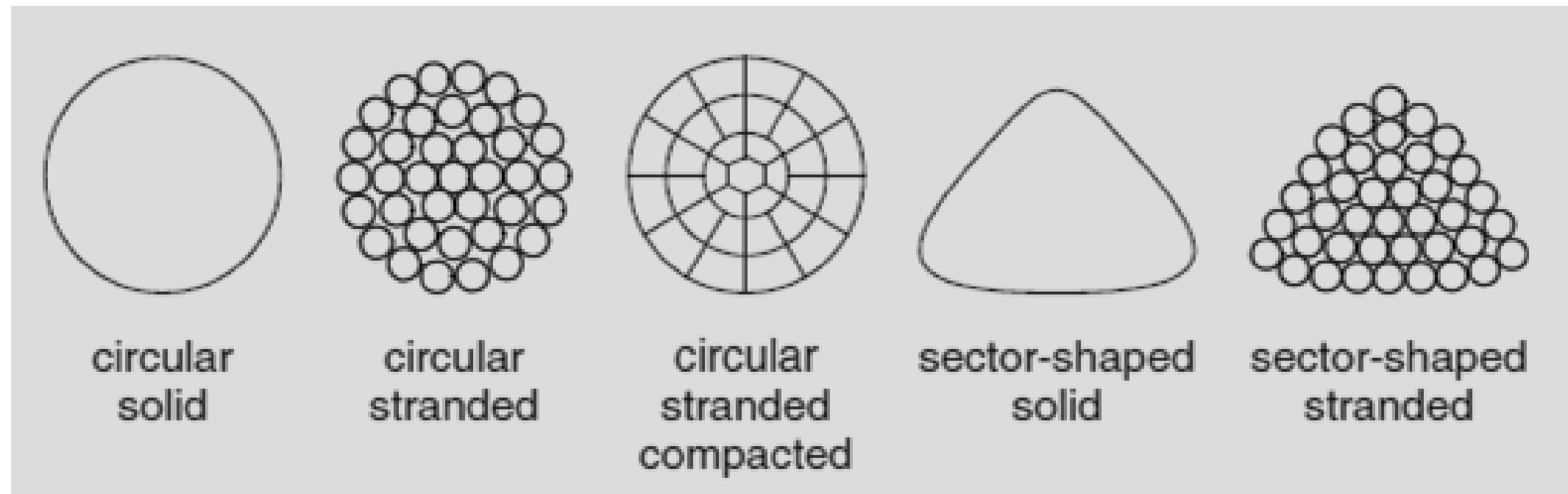
(a)



(b)

Figure 3.48 A three-core (a) and single-core (b) cable; (a) 6/10 kV with $3 \times 240 \text{ mm}^2$ aluminum, circular solid, conductors and XLPE insulation; (b) 220/380 kV with $1 \times 1600 \text{ mm}^2$ copper, circular stranded compacted, conductors and XLPE insulation. Reproduced by permission of Prysmian.

Conductor construction



- ◆ May contain several strands to fulfill flexibility specifications

Plastic insulation

- ◆ Polyethylene (PE) or cross-linked polyethylene (XLPE)
- ◆ Must be free of cavities and inclusions
- ◆ Must be sealed against water
- ◆ Copper strands in outer layer serve as armour

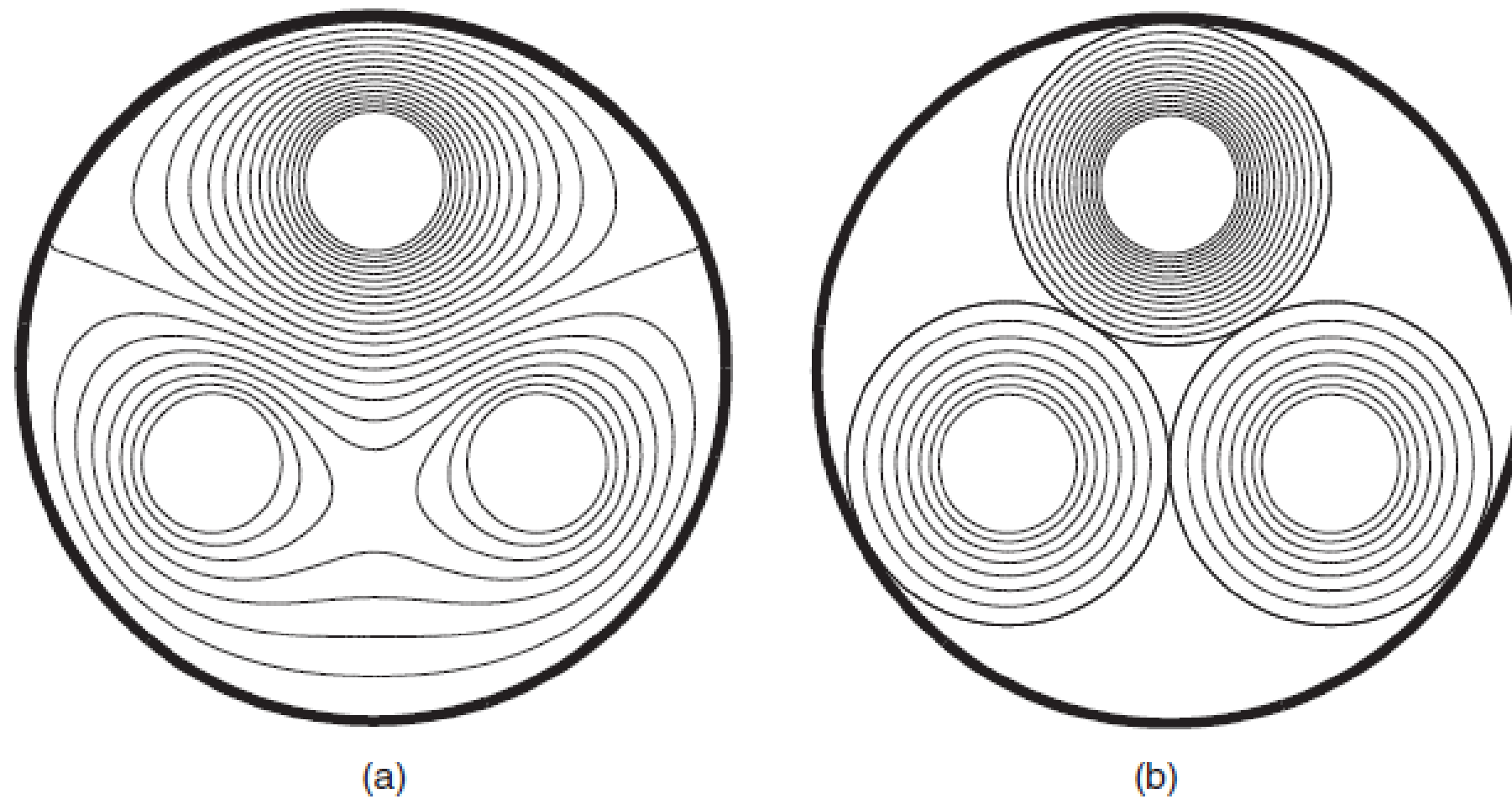


Paper-oil insulation

- ♦ Oil-impregnated paper
- ♦ Oil is stored in reservoirs along the cable
- ♦ Nowadays, mainly used for high and ultra-high voltage levels



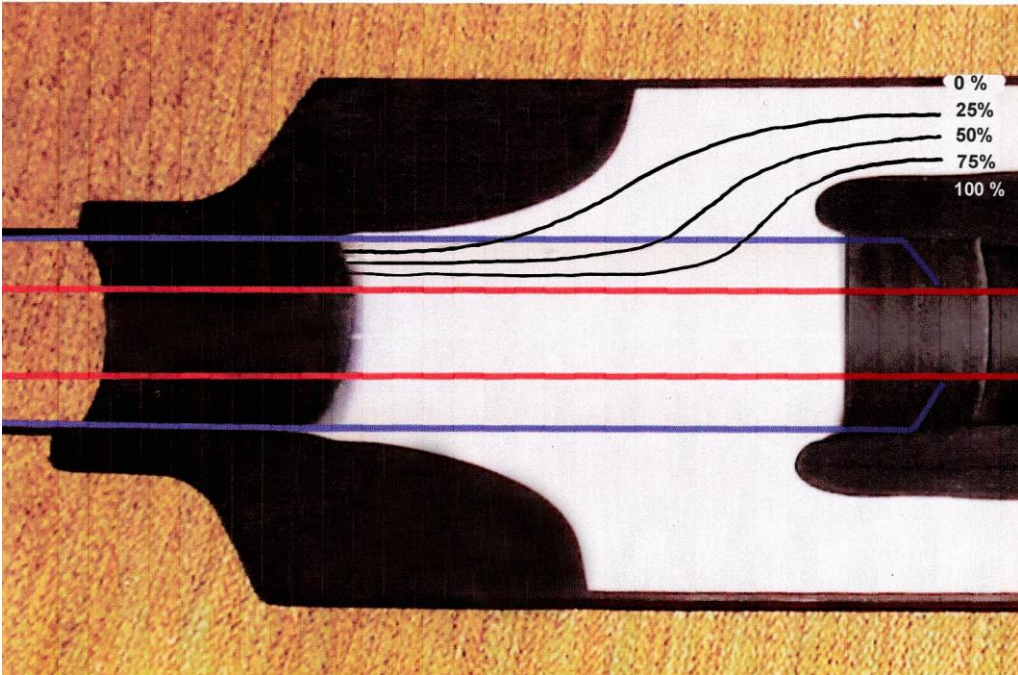
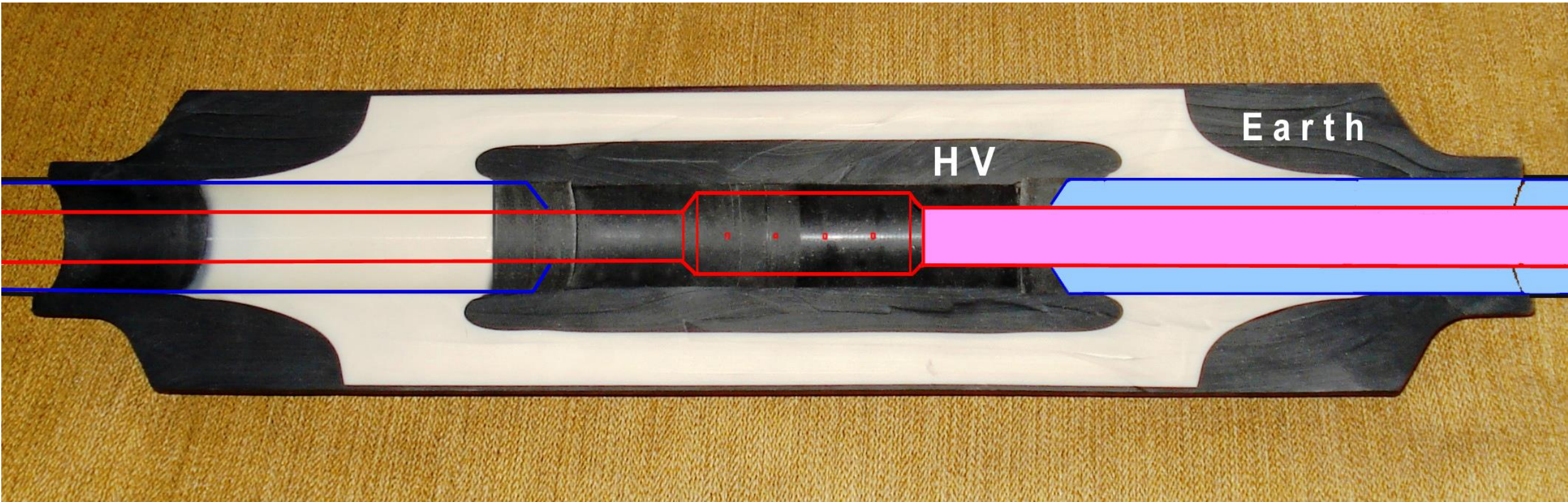
Electrical field



- ◆ Snapshot:
Phase A at
maximum positive
voltage,
Phase B and C at
half negative
voltage

Figure 3.49 Belted cable (a) and a three-core cable where each of the three phase conductors has its own sheath (b) and their equipotential lines.

Cable joints



Example: Charging current of an underground cable

Power carrier characteristics	Overhead line	Underground cable
Conductor material	copper	copper
Conductor area [mm ²]	150	240
Rated frequency [Hz]	50	50
Rated voltage [kV]	150	150
Rated power [MVA]	130	135
r [Ω /km]	0.125	0.12
x [Ω /km]	0.425	0.166
c [nF/km]	11.7	210

(Van der Sluis 2008)

- ◆ Approximation of cable capacitance (cylindrical capacitor):

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\frac{R}{r}}$$

Exercises



Exercise 1: Charging current of underground cable

- ◆ Consider the parameters of the 150 kV underground cable that are specified in Table 3.2.
 - a) Compute the maximum current that is allowed to flow in this cable.
 - b) Compute the current required to charge the distributed capacitance of the cable
 - c) Compute the length, at which the charging current at the sending end of the cable matches its maximum current.

Table 3.2 Comparison of typical parameters of a 150 kV overhead line and underground cable.

Power carrier characteristics	Overhead line	Underground cable
Conductor material	copper	copper
Conductor area [mm ²]	150	240
Rated frequency [Hz]	50	50
Rated voltage [kV]	150	150
Rated power [MVA]	130	135
r [Ω /km]	0.125	0.12
x [Ω /km]	0.425	0.166
c [nF/km]	11.7	210

References



References

- ♦ V. Crastan, *Elektrische Energieversorgung 1* (2nd Ed., Springer, 2010), pp. 169-224
- ♦ C. Harting, “[AC Transmission Line Losses](#)”, Submitted as coursework for *Physics 240* (Stanford University, 2010)
 - W. Hayt and J. Buck, *Engineering Electromagnetics* (McGraw-Hill, 2006), pp. 346, 486
 - F. W. Peek, “The Law of the Corona and the Dielectric Strength of Air”, *Transactions of A.I.E.E.* 30, 1889 (1911).
- ♦ Peter McLean, “[Lecture 5 – Transmission Line Parameters](#)”, *Power Circuit Theory* (University of Technology Sydney, 2012), p. 9
- ♦ B. M. Weedy [et al.], *Electric Power Systems* (5th Ed., Wiley, 2012), pp. 103-124
- ♦ L. Van der Sluis, P. Schavemaker, *Electrical Power System Essentials* (Wiley, 2008), pp. 98-118